

(NIT3004) IT Capstone Project 2

Block 2, Semester 2 2026

Submission Date:
28 August 2026



PROGRESS REPORT

LINGUISTAI

AI LANGUAGE LEARNING COMPANION


LinguistAI

Your AI Language Learning Companion

Team Members:

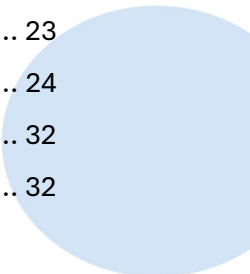
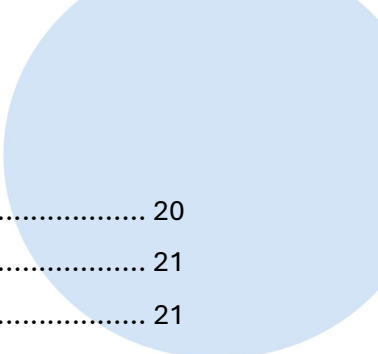
s8131699 - Quoc Anh, Nguyen
s8200193 - Htay Hlaing, Win
s8159573 – Camilo, Salazar Diaz
s8187769 - Jeison Andres, Aguirre Angulo

Block Convenor:

Dr Sardar Farhad

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LinguistAI

Your AI Language Learning Companion

1. Introduction.

As a group, we are developing **LinguistAI**, an interactive, web-based spoken language learning companion designed to help users bridge the "**fluency plateau**" through real-time, voice-to-voice conversation practice. Powered by Django, Supabase PostgreSQL, and multi-provider AI services (Google Gemini, AssemblyAI, and ElevenLabs), the platform delivers dynamic roleplay scenarios, natural AI speech synthesis, multi-layered feedback on grammar and pronunciation, and dynamic progress analytics.

The **LinguistAI** platform is currently 80% complete overall. Core backend pipelines, 3NF database models with Row Level Security (RLS), real-time speech processing, AI conversational brains, and dashboard analytics services are fully engineered and functional. The remaining scope involves hooking up specific dynamic frontend rendering widgets (such as the mispronounced words list) and finalizing live production hardening configurations. Our source code and setup guides are hosted on GitHub, all capstone documentation and project resources are organized in our team shared folders, and our implementation schedule is actively tracked on **Teamhood**:

GitHub Repository: <https://github.com/SympleGuy/LinguistAI>

Shared Folder (Capstone 1): [https://livevuedu-my.sharepoint.com/:f:/r/personal/s8187769_live_vu_edu_au/Documents/B4%20-%20\(NIT3003\)%20IT%20Capstone%20Project%201%20-%20SY?d=w9fe2dc158a5f4bbdac6a585b08d6efd2&csf=1&web=1&e=htKzw5](https://livevuedu-my.sharepoint.com/:f:/r/personal/s8187769_live_vu_edu_au/Documents/B4%20-%20(NIT3003)%20IT%20Capstone%20Project%201%20-%20SY?d=w9fe2dc158a5f4bbdac6a585b08d6efd2&csf=1&web=1&e=htKzw5)

Shared Folder (Capstone 2): [https://livevuedu-my.sharepoint.com/:f:/r/personal/s8187769_live_vu_edu_au/Documents/B2%20-%20\(NIT3004\)%20IT%20Capstone%20Project%202%20-%20SY?d=w37dca24b311b4053bf61f7c20baf2afe&csf=1&web=1&e=szlaKu](https://livevuedu-my.sharepoint.com/:f:/r/personal/s8187769_live_vu_edu_au/Documents/B2%20-%20(NIT3004)%20IT%20Capstone%20Project%202%20-%20SY?d=w37dca24b311b4053bf61f7c20baf2afe&csf=1&web=1&e=szlaKu)

Teamhood Project Tracking Gantt Chart:

<https://itcapstonegroup2.teamhood.com/VUSYD/Board/ITC2?view=GANT>

Note: For security reasons, please contact the project manager to request access to the links above.

2. Progress Overview.

As a group, we have reached an **80% overall completion milestone** for **LinguistAI**. The architecture, backend services, core databases, multi-provider AI integrations, and real-time voice handling are fully implemented and verified. The active progress of all project tasks is continuously stored and tracked on our **Teamhood Gantt Chart**.

2.1. Research and Planning.

Architecture & DB Design: Evaluated competitor platforms (e.g., Duolingo, Rosetta Stone) to address the "fluency plateau". Designed a 3NF relational schema in PostgreSQL/Supabase, incorporating Row Level Security (RLS) policies to ensure strict tenant isolation.

Cost & Latency Modeling: Analysed unit costs for external API usage (AssemblyAI, Google Gemini, ElevenLabs) to support a sustainable freemium model with rate limits.

2.2. Development.

Core Infrastructure & Auth (Phases 1 & 2): Configured the Django framework baseline, environment variables, Supabase connection, PBKDF2 password hashing, and dynamic CEFR profile selection (A1–C2).

Scenarios & Rate Limiting (Phase 3): Built the scenario library database model, populated system prompts, and implemented custom middleware enforcing a 5-turn/day limit for Free Tier accounts.

Voice Processing & AI Engine (Phases 4, 5 & 6): Integrated Web Audio API (**MediaRecorder**) for browser audio capture, AssemblyAI/Whisper for Speech-to-Text (STT), Google Gemini 1.5/2.0 Flash for dialogue generation/feedback, and ElevenLabs for Text-to-Speech (TTS). Implemented a Smart Fallback Simulation Engine in **myapp/ai_services.py** to handle missing API keys gracefully.

Analytics & Maintenance (Phases 7 & 8): Engineered server-rendered analytics endpoints with Chart.js visualization, B-Tree index query optimizations (**idx_sessions_user_id**), and an automated garbage collection script (**scripts/run_cleanup_cron.sh**) to purge temporary audio files older than 30 days.

2.3. Testing.

Unit & API Testing: Tested endpoints (**/api/sessions/start/**, **/api/sessions/<uuid>/respond-audio/**) and verified error-handling middleware to prevent stack trace leaks.

Local & Security Audits: Conducted cross-browser microphone checks and verified client-side authentication guards in **templates/linguistAi_web.html**. Enforced server-side execution of all external API calls so no credentials are exposed to the browser.

3. App Progress Report

3.1. Global Setup & Core Infrastructure Subsystem (Phase 1).

In this phase, we established the foundational Django project setup, configured environment variables, and integrated Supabase PostgreSQL.

Global Setup & Core Infrastructure Subsystem	
Files Used	
manage.py	
linguistai_project/settings.py	
linguistai_project/urls.py	
myapp/views.py	
templates/linguistAi_web.html	
.env	
requirements.txt	

Date	Team Member	Notes
22/06/2026	Quoc Anh Nguyen	Initialized Django project structure (manage.py , linguistai_project/), set up virtual environment dependencies, and committed requirements.txt
24/06/2026	Quoc Anh Nguyen	Configured environment variable management (.env) to securely handle Django secret keys and database URLs.
26/06/2026	Quoc Anh Nguyen	Configured core PostgreSQL database integration in linguistai_project/settings.py and executed initial migrations.
28/06/2026	Htay Hlaing Win	Created base HTML template (templates/linguistAi_web.html) and configured global Bootstrap 5 CSS/JS layout paths.
30/06/2026	Htay Hlaing Win	Implemented landing page rendering in myapp/views.py and mapped root routing in linguistai_project/urls.py .

Table 1 - Global Setup & Core Infrastructure Subsystem.

3.1.1. Home / Landing Page.

Displayed the LinguistAI brand title, main value proposition, and navigation bar featuring "Sign In" and "Get Started" buttons.

The screenshot displays the LinguistAI landing page. At the top, a navigation bar includes the LinguistAI logo, a 'Sign In' link, and a 'Try for Free' button. The main headline reads 'Practice Real Conversations, Not Flashcards.' with a sub-headline explaining that the platform uses AI to simulate real-life scenarios like ordering food or navigating airports. A 'Start Practising Free' button is prominently displayed, along with a 'View Demo' button. Below this, a section titled 'Everything you need to speak with confidence' lists six key features: Voice-First Practice, Real-Life Scenarios, Track Your Progress, Instant AI Feedback, Any Language, Any Level, and No Tutor Anxiety. A 'Learners who got there first' section features testimonials from three users: Sofia Müller, Karol Watanabe, and Amara Duffa. The 'Simple, honest pricing' section outlines three plans: a Free plan (\$0), a Monthly Premium plan (\$29.99), and a Lifetime Premium plan (\$19.99). The bottom of the page features a blue banner with the text 'Your first real conversation starts today.' and a 'Start Practising — It's Free' button.

Figure 1 - Home / Landing Page.

Displayed the running Django development server at <http://127.0.0.1:8000> alongside a log confirming a successful connection to the Supabase database.

The screenshot shows the Supabase dashboard's Logs section. The sidebar on the left contains navigation links for Logs, Time Range, Log Type, and Level. The main content area displays a table of log entries. The table has columns for DATE, METHOD, PATHNAME, and EVENT MESSAGE. A specific log entry is highlighted, showing a successful POST request to the network-bans endpoint. Below the table, the raw JSON of the log entry is displayed.

DATE	METHOD	PATHNAME	EVENT MESSAGE
26 Aug 26 13:11:14	00000		checkpoint complete: wrote 14 buffers (0.0%); 0 WAL file(s) added, 0 removed, 1 recycled; write
26 Aug 26 13:11:12	00000		checkpoint starting: time
26 Aug 26 13:10:45	00000		statement: SET statement_timeout='58s'; SET idle_session_timeout='58s'; CREATE OR REPLACE FUNCT
26 Aug 26 13:10:42	200	POST /admin/v1/network-bans/retrieve	POST 200 https://foxstplfvfqvqctmqm.supabase.co/admin/v1/network-bans/retrieve @supabase

```

RAW JSON
1 {
2   "id": "eebbe8ac-c875-4e4d-8dde-993d9a6ae36",
3   "date": "2026-08-26T03:10:42.127Z",
4   "method": "POST",
5   "pathname": "/admin/v1/network-bans/retrieve",
6   "status": "200",
7   "timestamp": "2026-08-26T03:10:42.127000",
8   "level": "success",
9   "event_message": "POST | 200 | https://foxstplfvfqvqctmqm.supabase.co/admin/v1/network-bans/retrieve | @supabase-infra/mgmt-api/7c73257",
10  "headers": {},
11  "regions": [],
12  "log_type": "edge",
13  "latency": 0,
14  "log_count": null,
15  "logs": [],
16  "auth_user": null
17 }

```

Figure 2 – Database Server.

A screenshot of a terminal window with a dark background and light green text. The terminal shows system information at the top, followed by directory navigation commands, and then the output of running a Django development server. The background of the terminal window features a blurred image of a street scene with people walking and Japanese signs.

```
13:10 Wed, Aug 26 /Projects/Lingus  
.....: .+++++: Terminal Font: HackNF-Regular (13pt)  
+++++:: ---- CPU: Intel(R) Core(TM) i5-8350U (8) @ 3.60 Ghz  
:*****-- GPU: Intel UHD Graphics 620 @ 1.10 Ghz [Integrated]  
:----- Memory: 1.85 GiB / 15.36 GiB (12%)  
:----- Swap: 3.10 MiB / 15.36 GiB (0%)  
:----- Disk (/): 53.25 GiB / 375.92 GiB (14%) - brfs  
:----- Local IP (wlan0): 172.16.96.25/16  
:----- Battery (01AV430): 79% [Discharging]  
:----- Locale: en_AU.UTF-8  
  
~  
> cd Projects/LinguistAI/  
  
~/Projects/LinguistAI main  
> /home/alex/venv/bin/python manage.py runserver 127.0.0.1:8000  
Watching for file changes with StatReloader  
Performing system checks...  
  
System check identified no issues (0 silenced).  
August 26, 2026 - 03:10:10  
Django version 4.2, using settings 'linguistai_project.settings'  
Starting development server at http://127.0.0.1:8000/  
Quit the server with CONTROL-C.
```

Figure 3 - Django Log.

3.2. Sign Up & Authentication Subsystem (Phase 2).

We implemented user registration, login, secure session cookies, target language settings, and dynamic CEFR proficiency profile configurations.

Sign Up & Authentication Subsystem
Files Used
myapp/models.py
myapp/views.py
myapp/middleware.py
templates/linguistAi_web.html
supabase_rls_policies.sql

Date	Team Member	Notes
01/07/2026	Quoc Anh Nguyen	Built the 3NF users database model in myapp/models.py and applied tenant security via supabase_rls_policies.sql .
03/07/2026	Htay Hlaing Win	Designed the authentication form modals inside templates/linguistAi_web.html with Bootstrap 5 styling.
05/07/2026	Htay Hlaing Win	Implemented front-end form validation to ensure email, username, and password fields are checked before submission.
06/07/2026	Camilo Salazar Diaz	Integrated target language and dynamic CEFR proficiency profile updates into myapp/views.py
07/07/2026	Quoc Anh Nguyen	Configured PBKDF2 password hashing, session authentication, and route security in myapp/middleware.py .

Table 2 - Sign Up & Authentication Subsystem.

3.2.1. Sign In / Registration Modal.

Showed the user onboarding modal designed for secure account access and profile setup.

The screenshot shows a web browser displaying the LinguistAI registration modal. The modal is titled "Create your account" and includes the following fields and options:

- Full Name:** A text input field with the placeholder "Your name".
- Email:** A text input field with the placeholder "you@example.com".
- Password:** A password input field with a strength indicator.
- Target Language to Learn:** A dropdown menu currently set to "French".
- Current Proficiency Level (CEFR):** A dropdown menu currently set to "Beginner (A1 - A2)".
- Create Account:** A blue button with a right-pointing arrow.
- Continue with Google:** A button with the Google logo.

The modal is highlighted with a red border. The background of the page shows a purple sidebar and a main content area with a quote: "I went from stumbling through French to ordering confidently in Paris – In just 6 weeks."

Figure 4 - Registration Modal.

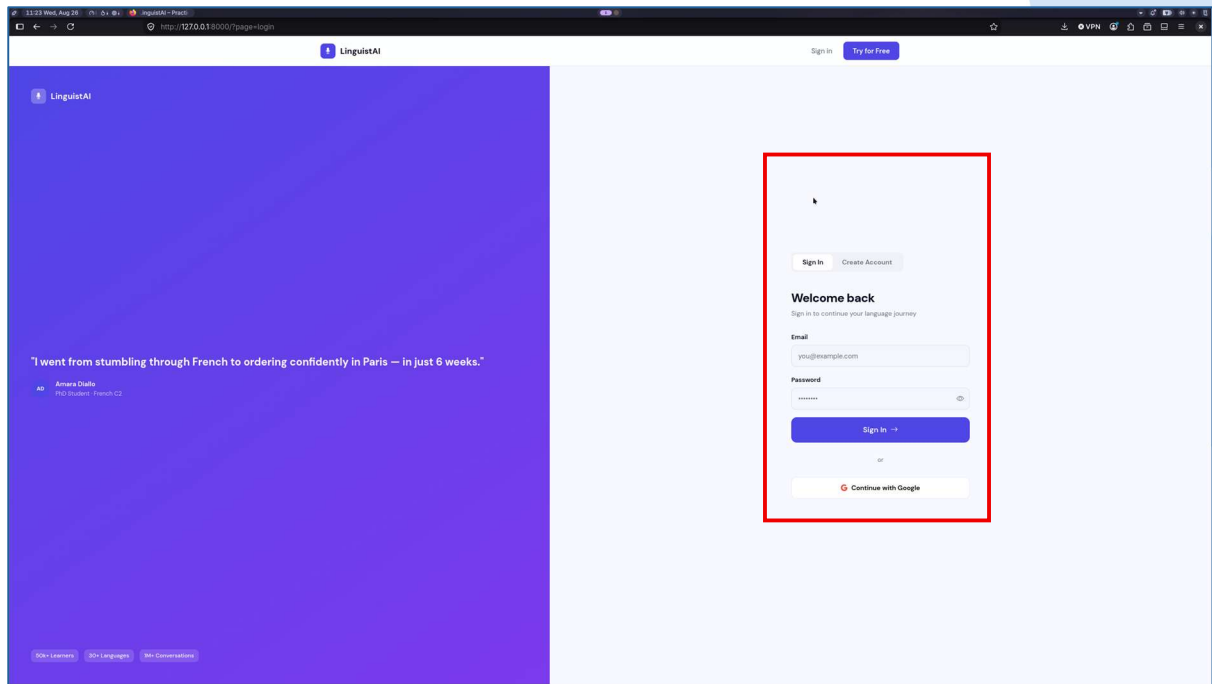


Figure 5 - Sign In Page.

3.2.2. Supabase Table Editor & RLS Policies.

Showed populated rows in the users table and verified active Row Level Security (RLS) policies.

id	username	password_hash	target_language	proficiency_level	subscription_plan
26ef4f60-d8e2-4b8a-aadc-0fbc23b97	Alex	pbkdf2_sha256\$600000\$wbktedKshF1	Japanese	Beginner	Free
4a8096eb-de1f-4308-b696-dc0cd7cd	Andres	pbkdf2_sha256\$600000\$BNIYgjH9x3b	German	Beginner	Free
cdabbc80-d96e-40e3-bb65-578b47d1	testuser	pbkdf2_sha256\$600000\$Mi6Nf9FHX7	French	Intermediate	Free
d242adbc-1ad0-4dad-b7b6-d00db507	test3	pbkdf2_sha256\$600000\$0KqRHwZdb	French	Beginner	Free
f3ca8396-daec-465c-a016-a5b088cd	Quoc Anh Nguyen	pbkdf2_sha256\$600000\$ypJlDXkNTZ	English	Beginner	Free

Figure 6 - Users Table.

NAME	COMMAND	APPLIED TO
Users can insert their own profile on signup	INSERT	public
Users can update their own profile	UPDATE	public
Users can view their own profile	SELECT	public

Figure 7 - Active Row Level Security (RLS) Policies.

3.3. Scenarios Library Subsystem (Phase 3).

We implemented the situational roleplay scenario catalog ("Ordering food," "Job interview"), integrated system prompt loading, and enforced free-tier daily usage limit middleware.

Scenarios Library Subsystem
Files Used
myapp/models.py
myapp/views.py
myapp/middleware.py
myapp/management/commands/seed_scenarios.py
templates/linguistAi_web.html

Date	Team Member	Notes
08/07/2026	Quoc Anh Nguyen	Created the scenarios database model in myapp/models.py to store roleplay contexts and system prompts.
10/07/2026	Quoc Anh Nguyen	Built and executed myapp/management/commands/seed_scenarios.py to seed default roleplay contexts (e.g., Restaurant, Airport, Job Interview).
11/07/2026	Htay Hlaing Win	Designed and rendered dynamic scenario selection UI cards grouped by CEFR level inside templates/linguistAi_web.html using Bootstrap 5.
12/07/2026	Htay Hlaing Win	Configured HTMX fragment loading in myapp/views.py to load the active chat room interface without full page refreshes.
13/07/2026	Jeison Andres Aguirre Angulo	Programmed rate-limiting middleware in myapp/middleware.py enforcing a 5-turn/day limit for Free Tier accounts (returning HTTP 403 Forbidden).

Table 3 - Scenarios Library Subsystem.

3.3.1. Scenarios Library View.

Showed the grid of interactive scenario cards (*Ordering Food, Job Interview*) categorized by CEFR proficiency levels.

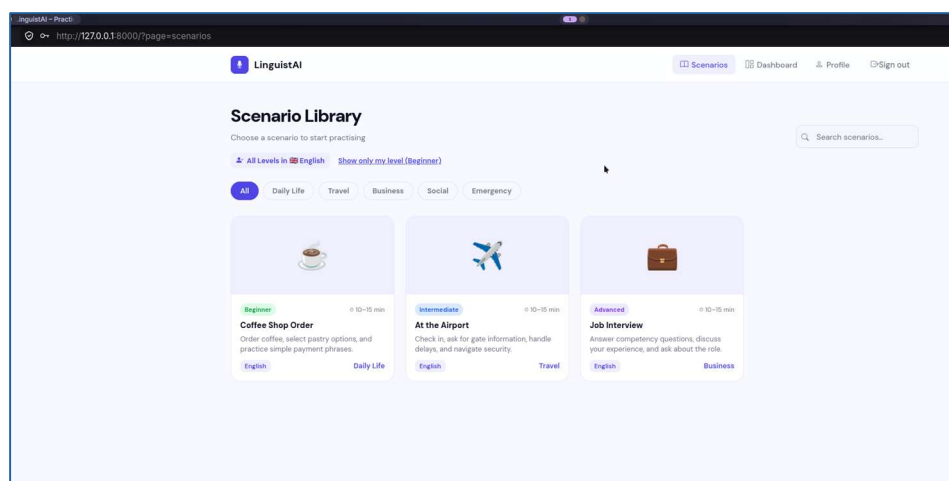


Figure 8 -Scenarios Library View.

3.3.2. Free-Tier Restriction Modal.

A popup triggered when a free user reaches the daily limit

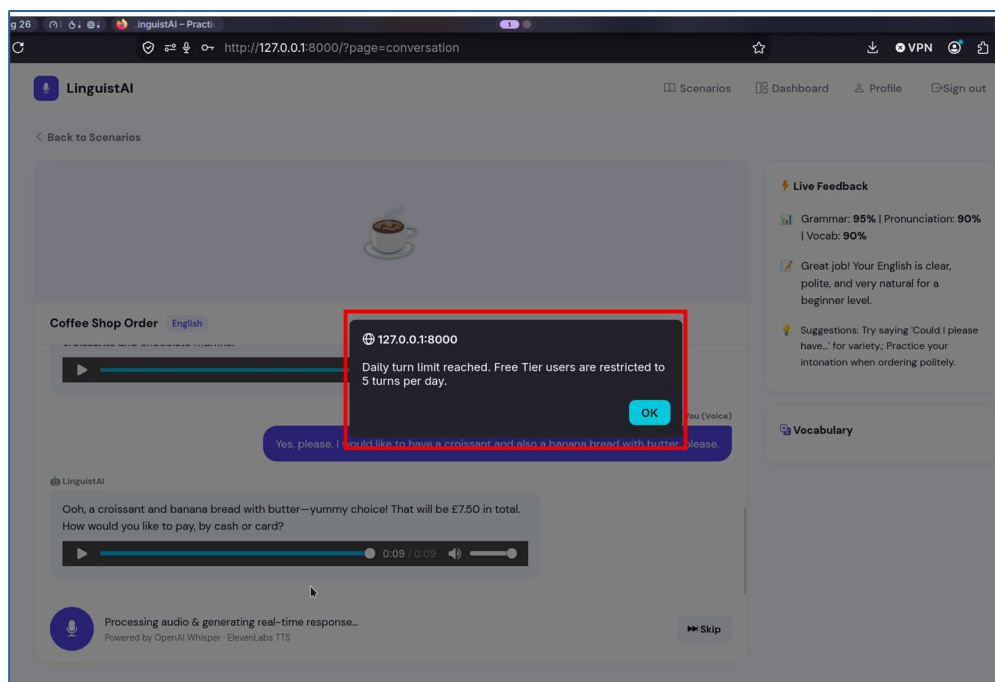


Figure 9 - Free-Tier Restriction Modal.

3.4. Audio Capture & Preparation Subsystem (Phase 4).

We integrated browser microphone APIs (**MediaRecorder**) for spoken input, constructed the HTMX chat shell, and implemented server-side audio verification.

Audio Capture & Preparation Subsystem	
Files Used	
myapp/models.py	
myapp/views.py	
templates/linguistAi_web.html	
linguistai_project/urls.py	

Date	Team Member	Notes
15/07/2026	Quoc Anh Nguyen	Created learning_sessions and interaction_logs 3NF database models in myapp/models.py to store voice interaction data.
17/07/2026	Htay Hlaing Win	Configured the HTMX chat room interface shell and dynamic session state rendering in templates/linguistAi_web.html .
19/07/2026	Htay Hlaing Win	Implemented front-end browser microphone capture using the Web Audio API (MediaRecorder) to package recorded audio blobs.
21/07/2026	Quoc Anh Nguyen	Built the /api/sessions/<uuid>/respond-audio/ endpoint in myapp/views.py to handle incoming audio payload verification and temporary file storage.

Table 4 - Audio Capture & Preparation Subsystem.

3.4.1. Active Chat Shell UI / Recording State.

Showed the real-time recording state with browser microphone capture and the "Stop & Send" action button rendered seamlessly via HTMX.

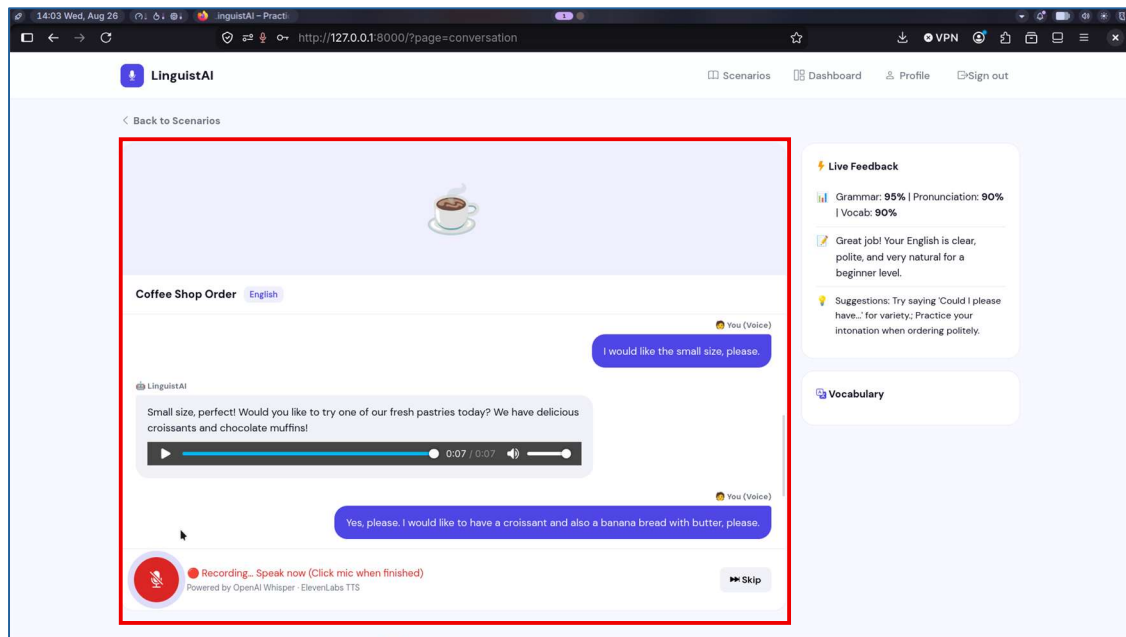


Figure 10 - Active Chat Shell UI / Recording State.

3.5. AI Brain & Conversation Core Subsystem (Phase 5).

We integrated Speech-to-Text (STT) transcription via AssemblyAI (Whisper fallback), established context-aware conversation memory retrieval, built dynamic LLM prompt assembly using Gemini (GPT-4 fallback), and integrated Text-to-Speech (TTS) voice generation via ElevenLabs.

AI Brain & Conversation Core Subsystem	
Files Used	
myapp/ai_services.py	
myapp/views.py	
myapp/middleware.py	
myapp/models.py	

Date	Team Member	Notes
22/07/2026	Camilo Salazar Diaz	Implemented backend logic in myapp/views.py to query historical turns from interaction_logs and build ordered conversation memory for context.
24/07/2026	Quoc Anh Nguyen	Integrated primary STT transcription via AssemblyAI API (with OpenAI Whisper secondary fallback) inside myapp/ai_services.py .
26/07/2026	Quoc Anh Nguyen	Built LLM prompt assembly and integrated Google Gemini 1.5/2.0 Flash (with OpenAI GPT-4o-mini fallback) to generate scenario responses and JSON evaluations.
28/07/2026	Camilo Salazar Diaz	Programmed the Smart Fallback Simulation Engine in myapp/ai_services.py and global exception middleware to handle API timeouts gracefully.

Table 5 - AI Brain & Conversation Core Subsystem.

3.5.1. Chat Thread - Transcripts & Conversational Responses.

Showed side-by-side conversational bubbles comparing the user's transcribed speech with the scenario-driven Gemini LLM responses.

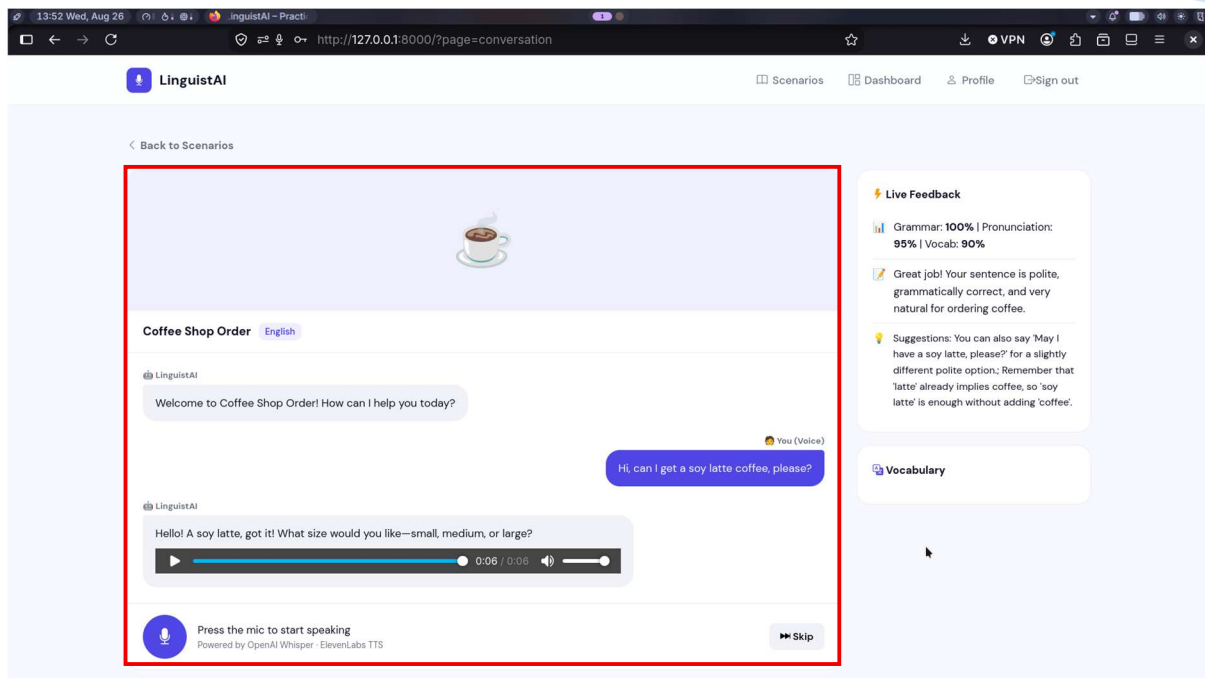


Figure 11 - Chat Thread - Transcripts & Conversational Responses.

3.5.2. DevTools: Django Development Server Logs for Spoken Response & AI Evaluation.

Browser Network tab showing the asynchronous STT API request payload (/api/sessions/<uuid>/respond-audio/) and response times.

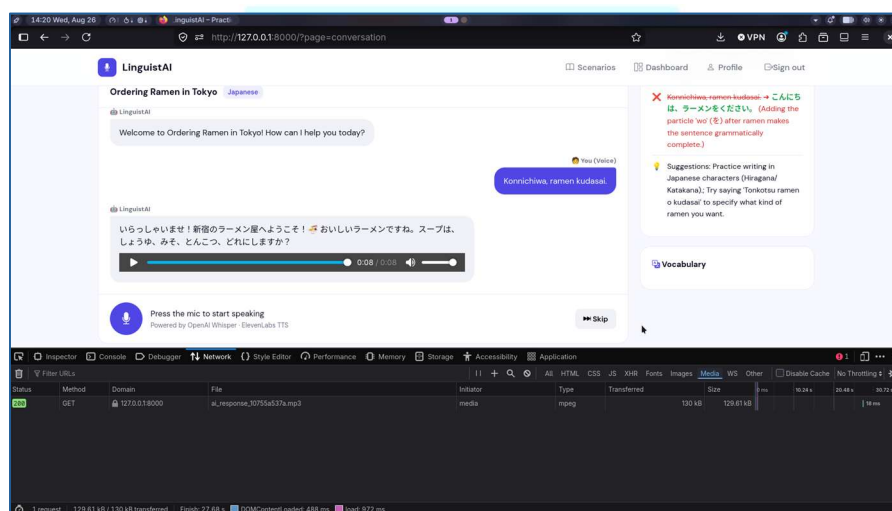


Figure 12 - Browser Network tab.

```
[26/Aug/2026 04:17:09] "GET /favicon.ico HTTP/1.1" 404 6421
[26/Aug/2026 04:17:10] "GET /api/auth/me/ HTTP/1.1" 200 221
[26/Aug/2026 04:17:12] "GET /api/scenarios/?lang=Japanese&cefr=Beginner&user_id=26ef4f60-d8e2-4b8a-aadc-0fbc23b979b9 HTTP/1.1" 200 460
[26/Aug/2026 04:17:15] "POST /api/sessions/start/ HTTP/1.1" 200 174
[26/Aug/2026 04:17:35] "POST /api/sessions/eade76a5-0775-40a0-81cd-645cf421475e/respond-audio/ HTTP/1.1" 200 1291
[26/Aug/2026 04:17:35] "GET /media/tts/ai_response_10755a537a.mp3 HTTP/1.1" 200 129611
```

Figure 13 - Response & AI Evaluation.

3.6. Voice & Conversational Flow Subsystem (Phase 6).

We integrated ElevenLabs Text-to-Speech (TTS) voice generation, delivered dynamic HTMX UI updates, and established JSONB logging for structured user feedback.

Voice & Conversational Flow Subsystem		
Files Used		
myapp/ai_services.py		
myapp/views.py		
templates/linguistAi_web.html		
myapp/models.py		

Date	Team Member	Notes
29/07/2026	Quoc Anh Nguyen	Integrated ElevenLabs TTS API into myapp/ai_services.py to synthesize realistic AI voice responses from generated text transcripts.
31/07/2026	Htay Hlaing Win	Configured HTML5 audio player rendering and dynamic audio source swapping inside templates/linguistAi_web.html .
02/08/2026	Htay Hlaing Win	Implemented dynamic HTMX response updates to insert newly generated audio players and transcript bubbles into the chat DOM without a full-page refresh.
04/08/2026	Camilo Salazar Diaz	Formatted JSON structured feedback (grammar corrections, pronunciation score, and suggestions) and logged records directly to interaction_logs in myapp/models.py .

Table 6 - Voice & Conversational Flow Subsystem.

3.6.1. Real-Time Feedback Panel & Audio Player.

Showed real-time grammar corrections, pronunciation metrics, and the auto-playing HTML5 audio control for the AI voice response.

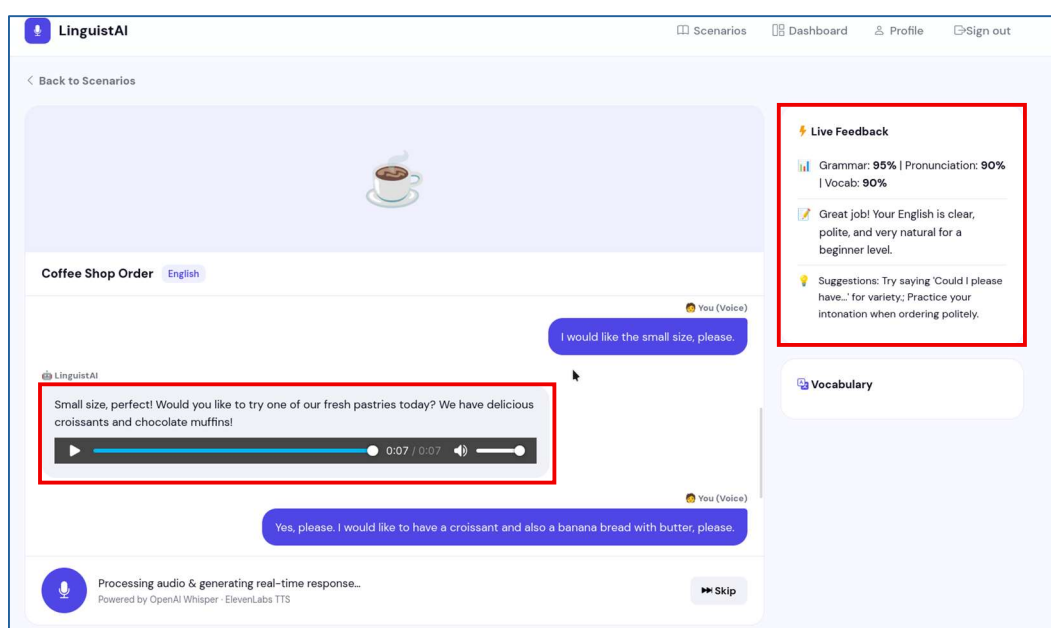
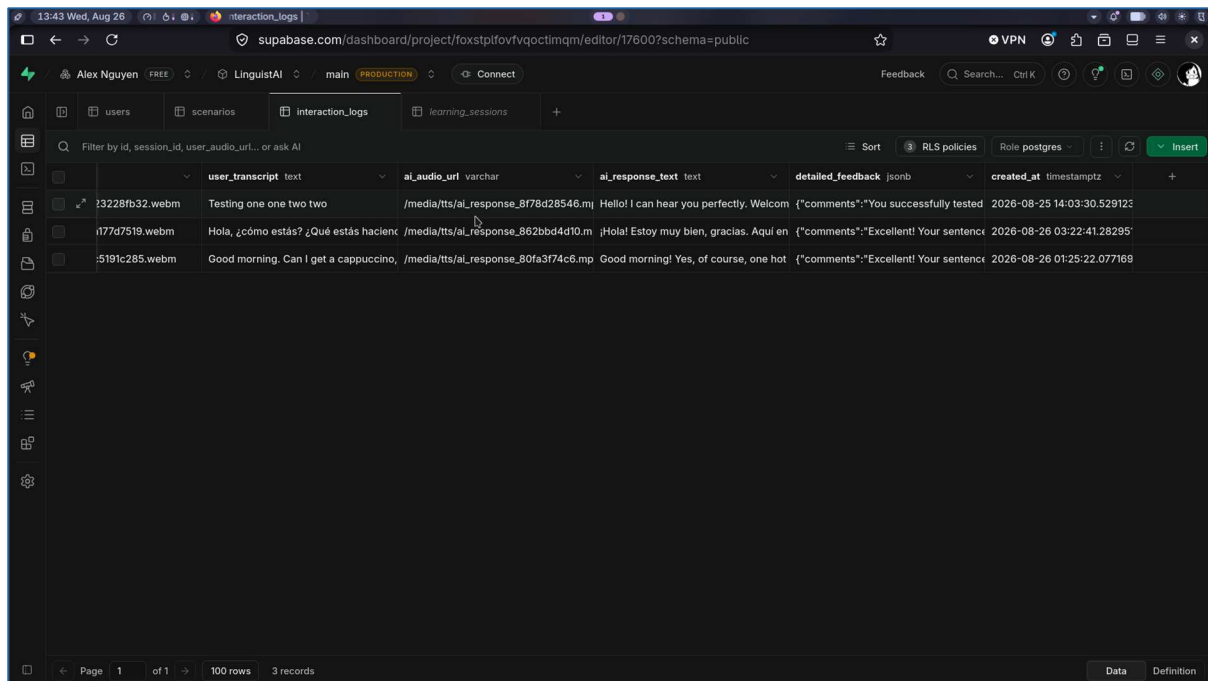


Figure 14 - Real-Time Feedback Panel & Audio Player.

3.6.2. Supabase Table Editor: Structured Feedback Logs

Supabase Table Editor showing a newly inserted row in **interaction_logs** with structured JSON feedback in the **detailed_feedback** column.



	user_transcript	ai_audio_url	ai_response_text	detailed_feedback	created_at
i3228fb32.webm	Testing one one two two	/media/tts/ai_response_8f78d28546.m	Hello! I can hear you perfectly. Welcom	("comments": "You successfully tested	2026-08-25 14:03:30.529123
i177d7519.webm	Hola, ¿cómo estás? ¿Qué estás hacien	/media/tts/ai_response_862bbd4d10.m	¡Hola! Estoy muy bien, gracias. Aquí en	("comments": "Excellent! Your sentence	2026-08-26 03:22:41.28295
i5191c285.webm	Good morning. Can I get a cappuccino.	/media/tts/ai_response_80fa3f74c6.m	Good morning! Yes, of course, one hot	("comments": "Excellent! Your sentence	2026-08-26 01:25:22.077169

Figure 15 - interaction_logs (Supabase)

Editing value of: detailed_feedback

```
1 {
2   "comments": "Excellent! Your sentences are grammatically perfect and very
3   natural for a beginner. Great job!",
4   "corrections": [],
5   "suggestions": [
6     "Try expanding your response by describing what you are doing, for
7     example: 'Estoy estudiando español' (I am studying Spanish).",
8     "Practice using alternative greetings like '¿Qué tal?' or '¿Cómo te va?'
9     to vary your vocabulary."
10  ],
11  "grammar_score": 100,
12  "vocabulary_score": 95,
13  "pronunciation_score": 95
14 }
```

Figure 16 - detailed_feedback (JSON Format)

3.7. Analytics & Dashboard Subsystem (Phase 7).

We computed key learning statistics, calculated consecutive day active streaks, and rendered interactive Chart.js progress charts alongside mispronounced word lists. (**Mispronounced word lists are still in work in progress**).

Analytics & Dashboard Subsystem	
Files Used	
myapp/views.py	
myapp/urls.py	
templates/linguistAi_web.html	
supabase_rls_policies.sql	

Date	Team Member	Notes
05/08/2026	Quoc Anh Nguyen	Applied database performance optimizations by creating B-Tree indexes (idx_sessions_user_id and idx_logs_session_id) to accelerate dashboard metrics aggregation.
07/08/2026	Jeison Andres Aguirre Angulo	Developed analytics views in myapp/views.py (/api/dashboard/<uuid>/ and /api/user/<uuid>/analytics/) to aggregate overall scores, calculate learning streaks, and generate Chart.js performance series.
10/08/2026	Htay Hlaing Win	Integrated front-end dashboard panels into templates/linguistAi_web.html rendering interactive Chart.js line/area charts, stat cards, and mispronounced word lists.

Table 7 - Analytics & Dashboard Subsystem.

3.7.1. User Progress Dashboard.

Showed Chart.js visualizations tracking weekly progression metrics paired with key stat cards for active streaks and overall accuracy.

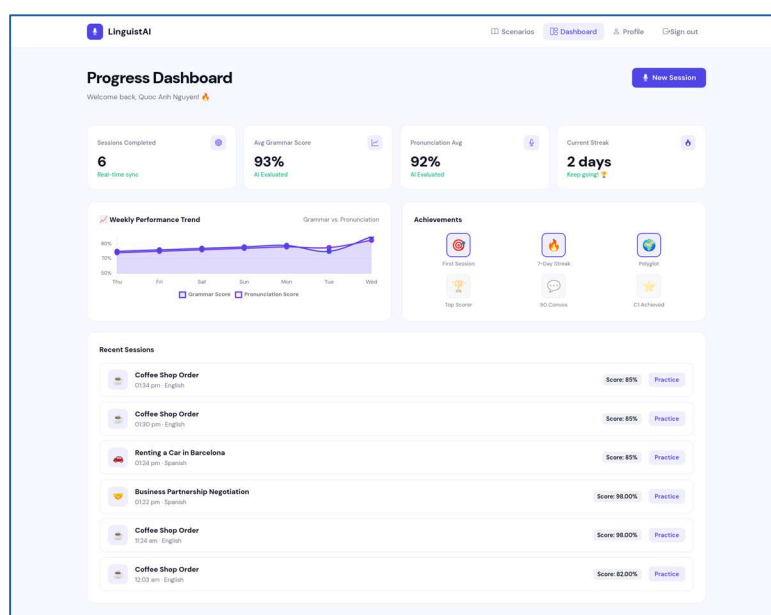


Figure 17 - User Progress Dashboard.

Mispronounced Words List: It is a dedicated card listing terms consistently flagged during speaking practice (**scheduled for the final implementation**).

3.8. System QA, Security & Maintenance Subsystem (Phase 8).

We executed rate-limiting security middleware, implemented B-Tree query optimizations, completed accessibility checks, and configured automated audio file cleanup CRON jobs. Additionally, we documented pending post-deployment security enhancements, including HTTPS enforcement, production CORS policies, and advanced rate-limiting controls.

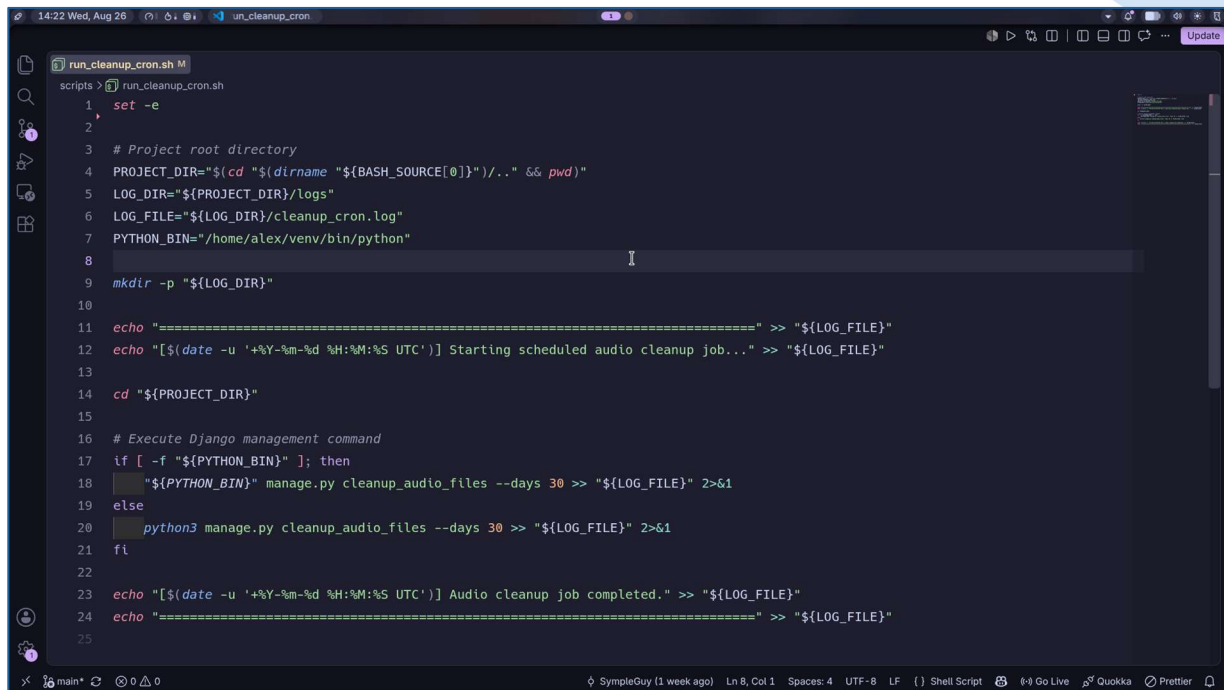
System QA, Security & Maintenance Subsystem (Phase 8)
Files Used
myapp/middleware.py
linguistai_project/settings.py
scripts/run_cleanup_cron.sh
templates/linguistAi_web.html

Date	Team Member	Notes
12/08/2026	Quoc Anh Nguyen	Executed local environment testing and configured debug logging flags in linguistai_project/settings.py .
15/08/2026	Jeison Andres Aguirre Angulo	Programmed basic exception-handling middleware in myapp/middleware.py to intercept API timeouts and fallback gracefully during local execution.
18/08/2026	Htay Hlaing Win	Conducted local UI responsiveness checks, fixed modal alignment issues, and verified cross-browser compatibility for browser audio recording in templates/linguistAi_web.html .
21/08/2026	Quoc Anh Nguyen	Created and tested scripts/run_cleanup_cron.sh to handle automated local garbage collection and delete temporary local audio files older than 30 days.
25/08/2026	Camilo Salazar Diaz	Documented pending security enhancements (such as HTTPS enforcement, production CORS policies, and advanced rate-limiting) for post-deployment production setup.

Table 8 - System QA, Security & Maintenance Subsystem.

3.8.1. CRON Script (scripts/run_cleanup_cron.sh)

For deleting audio files older than 30 days while preserving text transcripts.



```
1 set -e
2
3 # Project root directory
4 PROJECT_DIR="$(cd "$(dirname "${BASH_SOURCE[0]}")/.." && pwd)"
5 LOG_DIR="${PROJECT_DIR}/logs"
6 LOG_FILE="${LOG_DIR}/cleanup_cron.log"
7 PYTHON_BIN="/home/alex/venv/bin/python"
8
9 mkdir -p "${LOG_DIR}"
10
11 echo "===== " >> "${LOG_FILE}"
12 echo "[$(date -u '+%Y-%m-%d %H:%M:%S UTC')] Starting scheduled audio cleanup job..." >> "${LOG_FILE}"
13
14 cd "${PROJECT_DIR}"
15
16 # Execute Django management command
17 if [ -f "${PYTHON_BIN}" ]; then
18     "${PYTHON_BIN}" manage.py cleanup_audio_files --days 30 >> "${LOG_FILE}" 2>&1
19 else
20     python3 manage.py cleanup_audio_files --days 30 >> "${LOG_FILE}" 2>&1
21 fi
22
23 echo "[$(date -u '+%Y-%m-%d %H:%M:%S UTC')] Audio cleanup job completed." >> "${LOG_FILE}"
24 echo "===== " >> "${LOG_FILE}"
25
```

Figure 18 - CRON Script.

4. Design and Style Constants.

The design and style parameters for the **LinguistAI** project prioritize a modern UI, efficient DOM updates via HTMX, and complete server-side security.

4.1. Colour Palette & Typography.

Primary Palette: Built using Bootstrap 5 utility classes.

- **Brand Primary: #0D6EFD** (Bootstrap Blue) - Main actions, primary buttons, user speech bubbles.
- **Success/Accent: #198754** (Bootstrap Green) - Highlighting correct grammar phrasing, score badges.
- **Warning/Danger: #DC3545** (Bootstrap Red) & **#FFC107** (Bootstrap Yellow) - Flagging mispronounced tokens, grammatical errors, and daily turn usage limit alerts.
- **Background & Cards:** Light grey background (**#F8F9FA**) paired with white container cards (**#FFFFFF**) to ensure visual contrast and dark mode compatibility.
- **Typography:** Native system font stack via Bootstrap 5 (system-ui, -apple-system, Segoe UI, Roboto), ensuring fast loading and high readability across desktop and tablet viewports.

4.2. UI Architecture & Layout Specifications.

Layout Structure: Single Page Application (SPA) experience served via **templates/linguistAi_web.html** using a top navigation bar, fixed container cards, and responsive modal dialogs.

Chat Room Thread (/scenario/<id>):

- User speech bubbles are aligned right (Bootstrap blue background with white text).
- AI partner speech bubbles are aligned left (Light grey background with dark text).
- Async HTMX swaps inject text transcripts and HTML5 <audio> players independently to ensure audio synthesis does not block UI rendering.

Scenarios Library (/scenarios): Rendered as interactive Bootstrap cards displaying roleplay context details, system prompts, CEFR levels (A1–C2), and language badges.

Analytics Dashboard (/dashboard): Structured layout containing Stat Cards (Streak, Total Turns, Average Accuracy), Chart.js progress graphs, and a flagged terms list.

4.3. Styling & Component Rules

No Page Reloads: All UI updates from scenario loading to transcript swaps and feedback drawers use HTMX HTML fragment swaps.

Responsive Breakpoints: Fully functional across viewports 768px wide or larger (**Tablets and Desktops**) adhering to WCAG 2.1 Level AA accessibility standards.

Server-Only Credentials: Zero API keys or sensitive configurations exposed in CSS/JS client code; all styles load via local/CDN assets and authentication is handled using HTTP-only Django session cookies.

5. Database Schema.

The **LinguistAI** platform implements a normalized **3NF Relational Database Schema** hosted on PostgreSQL via Supabase. Primary keys use **UUIDs** across transactional tables to ensure tenant security and mitigate Insecure Direct Object Reference (IDOR) vulnerabilities.

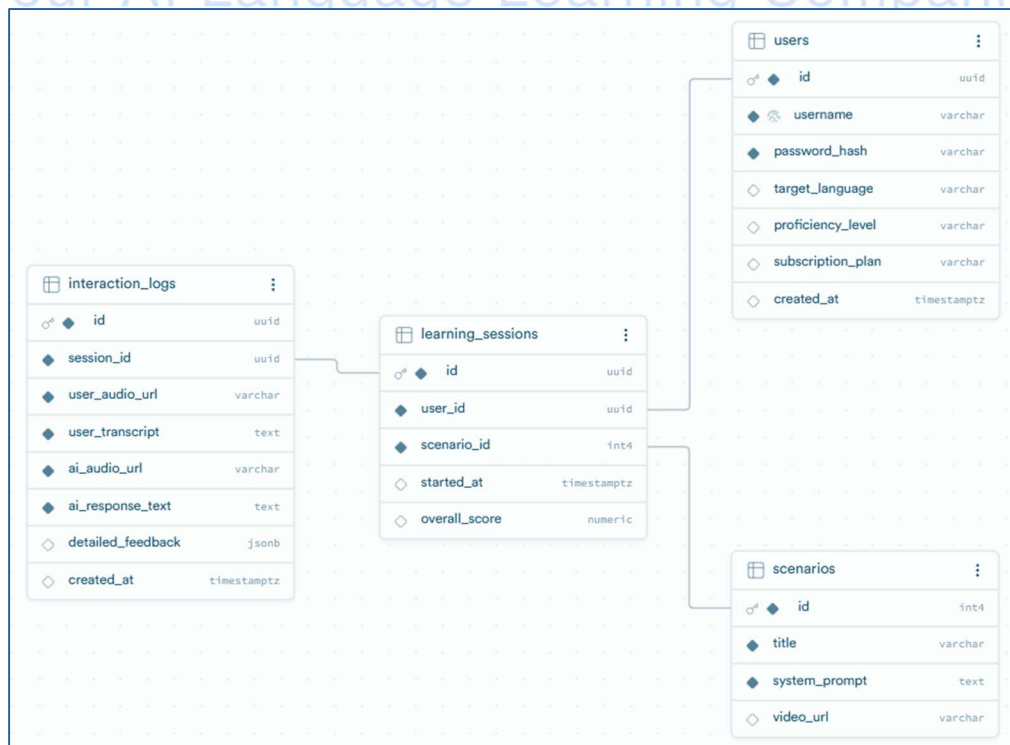


Figure 19 - Database Schema.

6. Remaining Work.

While the core architecture, multi-provider AI pipeline, database models, and primary user journeys for **LinguistAI** are fully functional, a concise set of final technical tasks remains to bring the platform to 100% completion and release readiness.

6.1. Final Features.

Dynamic Mispronounced Words UI Hook: The server-rendered analytics endpoint in `myapp/views.py` already aggregates user pronunciation metrics, but the dedicated UI widget in `templates/linguistAi_web.html` needs to be dynamically wired up to render the real-time list of flagged terms.

6.2. Final Testing and Bug Fixes.

Live Provider API Key Toggling: Transition the platform from the Smart Fallback Simulation Engine in `myapp/ai_services.py` to live production external API key configurations for AssemblyAI, Google Gemini, and ElevenLabs.

Production Environment Hardening: Perform final cross-browser microphone verification, enforce production HTTPS/TLS 1.3 security headers, and finalize production CORS policy configurations prior to public hosting.

6.3. Documentation Finalization.

Production Deployment Guide: Complete step-by-step instructions in docs/detailing production server setup, environment key provisioning, and automated CRON scheduling.

Handover Report: Finalize the Capstone 2 project report, user guide, system architecture handoff documentation for repository delivery, poster and video demonstration.

7. Timeline and Next Steps for Remaining Work.

Projected Timeline & Next Steps (31 August – 14 September 2026)

To bring **LinguistAI** from its current 80% completion state to 100% release readiness, the team will execute a structured **2-week** final sprint. Tasks are distributed directly according to each team member's technical domain and established project role.

7.1. Projected Timeline & Task Allocation.

Date Range	Phase / Task	Team Member	Project Role
31 Aug – 02 Sep	Dynamic UI Widget Integration: Wire up the front-end mispronounced words list widget in templates/linguistAi_web.html to dynamically display API metrics aggregated in myapp/views.py	Quoc Anh Nguyen	Backend & Database Developer
03 Sep – 05 Sep	Live Provider API Toggling: Transition myapp/ai_services.py from the Smart Fallback Simulation Engine to live production API key configurations for Google Gemini, AssemblyAI, and ElevenLabs.	Quoc Anh Nguyen	Backend & Database Developer
06 Sep – 08 Sep	Production Hardening & Audit Execution: Execute cross-browser microphone testing, verify production HTTPS/TLS 1.3 flags, inspect CORS policies, and run WCAG 2.1 AA accessibility scans.	Camilo Salazar Diaz	Front-end, Quality & NFR Lead.
09 Sep – 11 Sep	Documentation Finalization & Handover Report: Complete production deployment guides in docs/ , update environment setup documentation, and assemble the final Capstone 2 project report.	Jeison Andres Aguirre Angulo	Project Manager
12 Sep – 14 Sep	Final System Sanity Check & Repository Delivery: Run complete end-to-end user journey tests, confirm CRON script execution (scripts/run_cleanup_cron.sh), and deliver the final repository.	Active Capstone Team	Active Capstone Team

Table 9 - Projected Timeline & Task Allocation.

7.2. Timeline Schedule Gantt Chart.

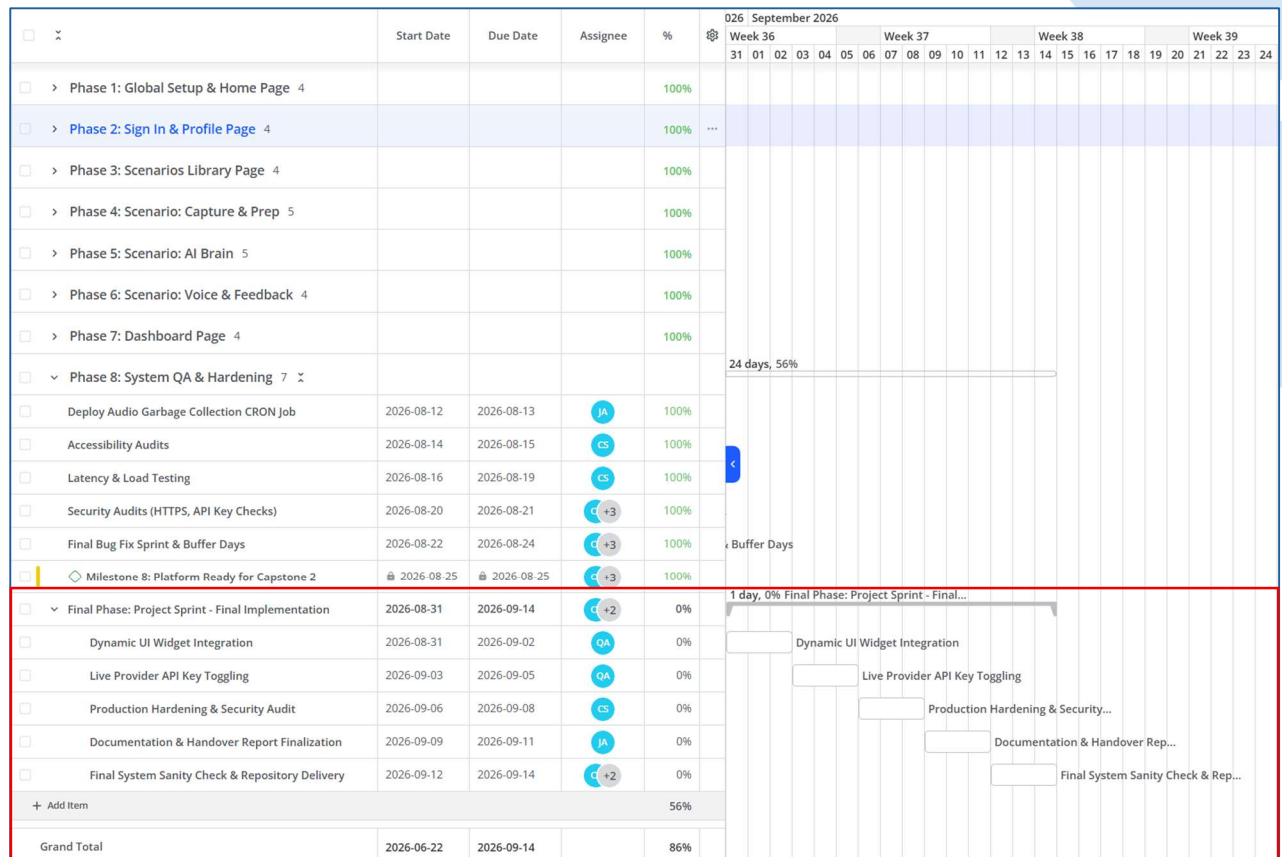


Figure 20 - Timeline Schedule Gantt Chart.

Your AI Language Learning Companion



9. Conclusion.

We have successfully driven the **LinguistAI** platform to an 80% completion milestone. The core backend architecture (**Django, PostgreSQL**), the real-time AI conversation pipeline (**AssemblyAI, Google Gemini, ElevenLabs**), and the dynamic database security protocols (**Supabase RLS**) are fully implemented and verified. These achievements mark a significant step toward bridging the "fluency plateau" by providing an affordable, 24/7 conversational environment that mirrors human interaction.

The remaining 20% of the project focuses on finalizing specific frontend visualizations specifically, hooking up the dynamic mispronounced words list widget in the dashboard and executing the final production hardening and security audits. Completing these final technical tasks over the coming two-week sprint is critical to transitioning the platform from a highly functional simulation to a robust, release-ready web application.

We invite questions and feedback from the audience regarding our implementation progress, the multi-provider AI pipeline, or our strategies for managing API fallback limits. Your insights will be invaluable as we refine LinguistAI and finalize our handover deliverables.



LinguistAI

Your AI Language Learning Companion

10. Meeting Minutes Report.

The Meeting minutes for all 8 project phases are attached below in chronological order:

MEETING MINUTES – PHASE 1

Project Title:	LinguistAI — AI Language Learning Companion		
Date of Meeting:	June 22, 2026	Time:	10:00 AM – 11:00 AM
Meeting Facilitator:	Andres Aguirre	Location:	Discord (Virtual)

1. Meeting Objective

Kick off Phase 1 of the implementation stage (Global Setup & Home Page), coordinate initial repository configuration, set up Supabase and Django infrastructure, and allocate initial tasks for the team.

2. Attendees

Name

Quoc Anh Nguyen - Backend & Database Developer

Htay Hlaing Win - Frontend & UI Developer

Camilo Salazar Diaz - Quality & NFR Lead

Jeison Andres Aguirre Angulo - Project Manager

3. Tasks Done Previous Week

Topic	Done by	Time
Finalized Capstone 1 Proposal & System Architecture	Jeison, Win, Camilo, Quoc Anh	June 14, 2026
Set up primary communication channels (Discord) & project tracking tools	Jeison Andres Aguirre Angulo	June 18, 2026
Prepared Gantt chart and implementation schedule	Quoc Anh Nguyen	June 20, 2026

4. Tasks to Be Done Next Week

Topic	To be done by	Time
Initialize Django project & Supabase database integration	Quoc Anh Nguyen	June 22 – June 25, 2026
Secure external API credentials (.env variables)	Quoc Anh Nguyen	June 26 – June 27, 2026
Develop Home Page layout using Bootstrap 5	Htay Hlaing Win	June 28 – June 29, 2026
Reach Milestone 1: Infrastructure & Home UI Functional	Htay Hlaing Win, Quoc Anh Nguyen	June 30, 2026

Submitted by: Jeison Andres Aguirre Angulo

MEETING MINUTES – PHASE 2

Project Title:	LinguistAI — AI Language Learning Companion		
Date of Meeting:	July 01, 2026	Time:	10:00 AM – 11:00 AM
Meeting Facilitator:	Andres Aguirre	Location:	Discord (Virtual)

1. Meeting Objective

Kick off Phase 2 of the project implementation (Sign In & Profile Page), set up the core PostgreSQL user database structure with Supabase Row Level Security (RLS) policies, plan the authentication system integration, and dynamic proficiency updates.

2. Attendees

Name

Quoc Anh Nguyen - Backend & Database Developer

Htay Hlaing Win - Frontend & UI Developer

Camilo Salazar Diaz - Quality & NFR Lead

Jeison Andres Aguirre Angulo - Project Manager

3. Tasks Done Previous Week

Topic	Done by	Time
Initialized Django project baseline & Supabase database integration	Quoc Anh Nguyen	June 22 – June 25, 2026
Configured external API credentials in .env variables	Quoc Anh Nguyen	June 26 – June 27, 2026
Developed Home Page UI layout using Bootstrap 5	Htay Hlaing Win	June 28 – June 29, 2026
Reached Milestone 1: Global Setup & Home Page functional	Htay Hlaing Win, Quoc Anh Nguyen	June 30, 2026

4. Tasks to Be Done Next Week

Topic	To be done by	Time
Build users database table & enforce Supabase RLS policies	Quoc Anh Nguyen	July 01 – July 02, 2026
Build Auth System & Sign In UI	Htay Hlaing Win	July 03 – July 05, 2026
Implement Dynamic CEFR Profile updates	Camilo Salazar Diaz	July 06, 2026
Reach Milestone 2: User Access & Profiles Secure	All Team Members	July 07, 2026

Submitted by: Jeison Andres Aguirre Angulo

MEETING MINUTES – PHASE 3

Project Title:	LinguistAI — AI Language Learning Companion		
Date of Meeting:	July 08, 2026	Time:	10:00 AM – 11:00 AM
Meeting Facilitator:	Andres Aguirre	Location:	Discord (Virtual)

1. Meeting Objective

Kick off Phase 3 of the implementation stage (Scenarios Library Page), coordinate the setup of the scenarios database table and prompt population, plan the development of scenario selection cards in the UI, and establish the free-tier usage restriction logic.

2. Attendees

Name

Quoc Anh Nguyen - Backend & Database Developer

Htay Hlaing Win - Frontend & UI Developer

Camilo Salazar Diaz - Quality & NFR Lead

Jeison Andres Aguirre Angulo - Project Manager

3. Tasks Done Previous Week

Topic	Done by	Time
<i>Built users DB table & enforced Supabase RLS policies</i>	Quoc Anh Nguyen	July 01 – July 02, 2026
<i>Built Auth System & Sign In UI</i>	Htay Hlaing Win	July 03 – July 05, 2026
<i>Implemented Dynamic CEFR Profile updates</i>	Camilo Salazar Dia	July 06, 2026
<i>Reached Milestone 2: User Access & Profiles Secure</i>	All Team Members	July 07, 2026

4. Tasks to Be Done Next Week

Topic	To be done by	Time
<i>Create scenarios DB table & populate default AI prompts</i>	Quoc Anh Nguyen	July 08 – July 10, 2026
<i>Render Scenario Selection UI cards using Bootstrap 5 & HTMX</i>	Htay Hlaing Win	July 11 – July 12, 2026
<i>Write usage limit logic (5-turns/day restriction for Free Tier)</i>	Jeison Andres Aguirre Angulo	July 13, 2026
<i>Reach Milestone 3: Scenarios Library & Limits Active</i>	All Team Members	July 14, 2026

Submitted by: Jeison Andres Aguirre Angulo

MEETING MINUTES – PHASE 4

Project Title:	LinguistAI — AI Language Learning Companion		
Date of Meeting:	July 15, 2026	Time:	10:00 AM – 11:00 AM
Meeting Facilitator:	Andres Aguirre	Location:	Discord (Virtual)

1. Meeting Objective

Kick off Phase 4 of the implementation stage (Scenario: Capture & Prep), coordinate the setup of speech processing tables (`learning_sessions` and `interaction_logs`), build the HTMX chat shell, implement frontend audio recording using the MediaRecorder API, and establish server-side audio verification.

2. Attendees

Name

Quoc Anh Nguyen - Backend & Database Developer

Htay Hlaing Win - Frontend & UI Developer

Camilo Salazar Diaz - Quality & NFR Lead

Jeison Andres Aguirre Angulo - Project Manager

3. Tasks Done Previous Week

Topic	Done by	Time
Created scenarios DB table & populated default AI prompts	Quoc Anh Nguyen	July 08 – July 10, 2026
Rendered Scenario Selection UI cards using Bootstrap 5 & HTMX	Htay Hlaing Win	July 11 – July 12, 2026
Written free-tier usage limit logic (5 turns/day limit)	Jeison Andres Aguirre Angulo	July 13, 2026
Reached Milestone 3: Scenarios Library & Limits Active	All Team Members	July 14, 2026

4. Tasks to Be Done Next Week

Topic	To be done by	Time
Create <code>learning_sessions</code> and <code>interaction_logs</code> database tables	Quoc Anh Nguyen	July 15 – July 16, 2026
Setup HTMX chat interface shell	Htay Hlaing Win	July 17 – July 18, 2026
Implement browser microphone capture via MediaRecorder API	Htay Hlaing Win	July 19 – July 20, 2026
Build server-side audio verification with pydub/librosa	Quoc Anh Nguyen	July 21, 2026
Reach Milestone 4: Audio Capture Pipeline Functional	Quoc Anh Nguyen, Htay Hlaing Win	July 21, 2026

Submitted by: Jeison Andres Aguirre Angulo

MEETING MINUTES – PHASE 5

Project Title:	LinguistAI — AI Language Learning Companion		
Date of Meeting:	July 22, 2026	Time:	10:00 AM – 11:00 AM
Meeting Facilitator:	Andres Aguirre	Location:	Discord (Virtual)

1. Meeting Objective

Kick off Phase 5 of the implementation stage (Scenario: AI Brain), establish conversation memory querying, integrate OpenAI Whisper STT and LLM prompt assembly (Claude/GPT-4), and set up cross-testing with fallback error handling.

2. Attendees

Name

Quoc Anh Nguyen - Backend & Database Developer

Htay Hlaing Win - Frontend & UI Developer

Camilo Salazar Diaz - Quality & NFR Lead

Jeison Andres Aguirre Angulo - Project Manager

3. Tasks Done Previous Week

Topic	Done by	Time
<i>Created learning_sessions and interaction_logs database tables.</i>	Quoc Anh Nguyen	July 15 – July 16, 2026
<i>Setup HTMX chat interface shell</i>	Htay Hlaing Win	July 17 – July 18, 2026
<i>Implemented browser microphone capture via MediaRecorder API</i>	Htay Hlaing Win	July 19 – July 20, 2026
<i>Built server-side audio verification with pydub/librosa</i>	Quoc Anh Nguyen	July 21, 2026
<i>Reached Milestone 4: Audio Capture Pipeline Functional</i>	Quoc Anh Nguyen, Htay Hlaing Win	July 21, 2026

4. Tasks to Be Done Next Week

Topic	To be done by	Time
<i>Build Conversation Memory Query logic</i>	Camilo Salazar Diaz	July 22 – July 23, 2026
<i>Integrate Speech-to-Text (STT) API via OpenAI Whisper</i>	Quoc Anh Nguyen	July 24 – July 25, 2026
<i>Assemble and integrate LLM prompts (Claude/GPT-4)</i>	Quoc Anh Nguyen	July 26 – July 27, 2026
<i>Conduct cross-testing and program fallback error messages</i>	Camilo Salazar Diaz	July 28, 2026
<i>Reach Milestone 5: Core AI Context & Responses Ready</i>	Quoc Anh Nguyen, Camilo Salazar Diaz	July 28, 2026

Submitted by: Jeison Andres Aguirre Angulo

MEETING MINUTES – PHASE 6

Project Title:	LinguistAI — AI Language Learning Companion		
Date of Meeting:	July 29, 2026	Time:	10:00 AM – 11:00 AM
Meeting Facilitator:	Andres Aguirre	Location:	Discord (Virtual)

1. Meeting Objective

Kick off Phase 6 of the implementation stage (Scenario: Voice & Conversational Flow), coordinate ElevenLabs Text-to-Speech (TTS) API integration, implement dynamic HTMX UI updates for real-time feedback (grammar/pronunciation), and set up JSONB storage for complete interaction logs.

2. Attendees

Name

Quoc Anh Nguyen - Backend & Database Developer

Htay Hlaing Win - Frontend & UI Developer

Camilo Salazar Diaz - Quality & NFR Lead

Jeison Andres Aguirre Angulo - Project Manager

3. Tasks Done Previous Week

Topic	Done by	Time
<i>Build Conversation Memory Query logic</i>	Camilo Salazar Diaz	July 22 – July 23, 2026
STT & LLM Migration: Replaced standalone Whisper with AssemblyAI (STT) and Google Gemini 1.5/2.0 Flash (LLM & Evaluation)	Quoc Anh Nguyen	July 24 – July 25, 2026
Assembled LLM prompts & implemented multi-provider fallback engine (<i>myapp/ai_services.py</i>)	Quoc Anh Nguyen	July 26 – July 27, 2026
Conduct cross-testing and program fallback error messages	Camilo Salazar Diaz	July 28, 2026
Reach Milestone 5: Core AI Context & Responses Ready	Quoc Anh Nguyen, Camilo Salazar Diaz	July 28, 2026

4. Tasks to Be Done Next Week

Topic	To be done by	Time
<i>Integrate Text-to-Speech (TTS) API via ElevenLabs</i>	Quoc Anh Nguyen	July 29 – July 30, 2026
<i>Implement dynamic HTMX fragment injections for grammar & pronunciation UI</i>	Htay Hlaing Win	July 31 – August 02, 2026
<i>Securely store detailed interaction payload in PostgreSQL JSONB column</i>	Quoc Anh Nguyen	August 03, 2026
Reach Milestone 6: Real-time Conversational Flow Complete	All Team Members	August 04, 2026

Submitted by: Jeison Andres Aguirre Angulo

MEETING MINUTES – PHASE 7

Project Title:	LinguistAI — AI Language Learning Companion		
Date of Meeting:	August 05, 2026	Time:	10:00 AM – 11:00 AM
Meeting Facilitator:	Andres Aguirre	Location:	Discord (Virtual)

1. Meeting Objective

Kick off Phase 7 of the implementation stage (Dashboard Page), implement B-Tree index optimizations on database queries, develop server-rendered dashboard analytics, and render the user progress UI and mispronounced word lists.

2. Attendees

Name

Quoc Anh Nguyen - Backend & Database Developer

Htay Hlaing Win - Frontend & UI Developer

Camilo Salazar Diaz - Quality & NFR Lead

Jeison Andres Aguirre Angulo - Project Manager

3. Tasks Done Previous Week

Topic	Done by	Time
Integrate Text-to-Speech (TTS) API via ElevenLabs	Quoc Anh Nguyen	July 29 – July 30, 2026
Implement dynamic HTMX fragment injections for grammar & pronunciation UI	Htay Hlaing Win	July 31 – August 02, 2026
Securely store detailed interaction payload in PostgreSQL JSONB column	Quoc Anh Nguyen	August 03, 2026
Reach Milestone 6: Real-time Conversational Flow Complete	All Team Members	August 04, 2026

4. Tasks to Be Done Next Week

Topic	To be done by	Time
Apply B-Tree index optimization for database queries (<i>idx_sessions_user_id</i> , <i>idx_logs_session_id</i>).	Quoc Anh Nguyen	August 05 – August 06, 2026
Develop server-rendered dashboard analytics	Jeison Andres Aguirre Angulo	August 07 – August 09, 2026
Render progress UI & mispronounced word lists	Htay Hlaing Win	August 10, 2026
Reach Milestone 7: Analytics & Dashboard Functional	Jeison Andres Aguirre Angulo, Htay Hlaing Win	August 11, 2026

Submitted by: Jeison Andres Aguirre Angulo

MEETING MINUTES – PHASE 8

Project Title:	LinguistAI — AI Language Learning Companion		
Date of Meeting:	August 12, 2026	Time:	10:00 AM – 11:00 AM
Meeting Facilitator:	Andres Aguirre	Location:	Discord (Virtual)

1. Meeting Objective

Kick off Phase 8 of the implementation stage (System QA & Hardening), set up scheduled automated audio file garbage collection CRON jobs, execute WCAG 2.1 AA accessibility audits, run latency and system load testing, perform security audits (HTTPS, TLS 1.3, API key checks), and execute the final bug fix sprint to reach releasing readiness.

2. Attendees

Name

Quoc Anh Nguyen - Backend & Database Developer

Htay Hlaing Win - Frontend & UI Developer

Camilo Salazar Diaz - Quality & NFR Lead

Jeison Andres Aguirre Angulo - Project Manager

3. Tasks Done Previous Week

Topic	Done by	Time
Apply B-Tree index optimization for database queries (<i>idx_sessions_user_id</i> , <i>idx_logs_session_id</i>).	Quoc Anh Nguyen	August 05 – August 06, 2026
Develop server-rendered dashboard analytics	Jeison Andres Aguirre Angulo	August 07 – August 09, 2026
Render progress UI & mispronounced word lists	Htay Hlaing Win	August 10, 2026
Reach Milestone 7: Analytics & Dashboard Functional	Jeison Andres Aguirre Angulo, Htay Hlaing Win	August 11, 2026

4. Tasks to Be Done Next Week

Topic	To be done by	Time
Deploy Audio Garbage Collection CRON job script	Jeison Andres Aguirre Angulo	August 12 – August 13, 2026
Conduct accessibility audits using axe-core (WCAG 2.1 AA compliance)	Camilo Salazar Diaz	August 14 – August 15, 2026
Perform latency & load testing via Locust	Camilo Salazar Diaz	August 16 – August 19, 2026
Run security audits (HTTPS/TLS 1.3, environment variable API key checks)	All Team Members	August 20 – August 21, 2026
Final bug fix sprint & buffer execution	All Team Members	August 22 – August 24, 2026
Reach Milestone 8: Platform Ready for Capstone 2 to complete 20% left.	All Team Members	August 25, 2026

Submitted by: Jeison Andres Aguirre Angulo

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